

Paper Reading Summaries

Prepared for capstone / literature review reference — privacy, informed consent & dark patterns

Paper 5: Circumvention by Design — Dark Patterns in Cookie Consents for Online News Outlets

Soe, T. H., Nordberg, O. E., Guribye, F., & Slavkovik, M. (2020). University of Bergen, Norway. *arXiv:2006.13985*

Overview

This paper investigates how online news outlets implement cookie consent notices required under the EU's General Data Protection Regulation (GDPR). The authors manually examined 300 real-world consent pop-ups from Scandinavian and English-language news websites to determine whether these notices genuinely support informed consent or instead use manipulative design ('dark patterns') to nudge users toward accepting data collection.

Methodology

- 300 cookie consent notices manually collected in July 2019 (250 news outlets, 50 magazines) across 42 countries.
- Each site revisited in April 2020 (3 months after CCPA took effect) to check for design changes. ● Two independent reviewers coded the presence of dark patterns using Gray et al.'s (2018) five-category taxonomy: Nagging, Obstruction, Sneaking, Interface Interference, Forced Action.
- Data included screenshots, consent text, and click-count required to deny consent. **Key Findings**
- 297 of 300 (99%) websites used at least one dark pattern when eliciting consent.
- Obstruction (43%) and Interface Interference (45.3%) were the most common dark pattern types.
- 51% of sites required 10-12 clicks to deny all cookies, while accepting took only 1 click.
- Only 15 of 300 sites offered a direct 1-click 'deny' option.
- Inter-rater reliability for classifying dark patterns was only 67%, showing Gray et al.'s categories are too broad/ambiguous for this specific context.
- 121 different wordings were used across sites for the 'deny' option, and 201 distinct phrasings for cookie purposes — showing no industry-wide standard vocabulary.

Proposed Contribution: 8 New Dark Pattern Types

Because Gray et al.'s five general categories proved too broad for cookie consent specifically, the authors proposed eight refined, context-specific dark pattern types:

- Does Not Count
- No Choice
- Multiple Choice Panels
- Choice Cascade

- Widget Inequality
- Unlabeled Sliders
- Unmarked X ● No Antonyms

Conclusion / Takeaway

Legal regulation (GDPR/CCPA) alone is insufficient to guarantee informed consent — the interface design through which consent is elicited must also be regulated. The authors argue that dark pattern definitions need to be precise and computer-detectable so regulators can eventually automate compliance checking. Future work proposed: build an automated crawler / ML classifier trained on their labeled 300-site dataset to detect these patterns at scale.